01 / RMS COMPARISON

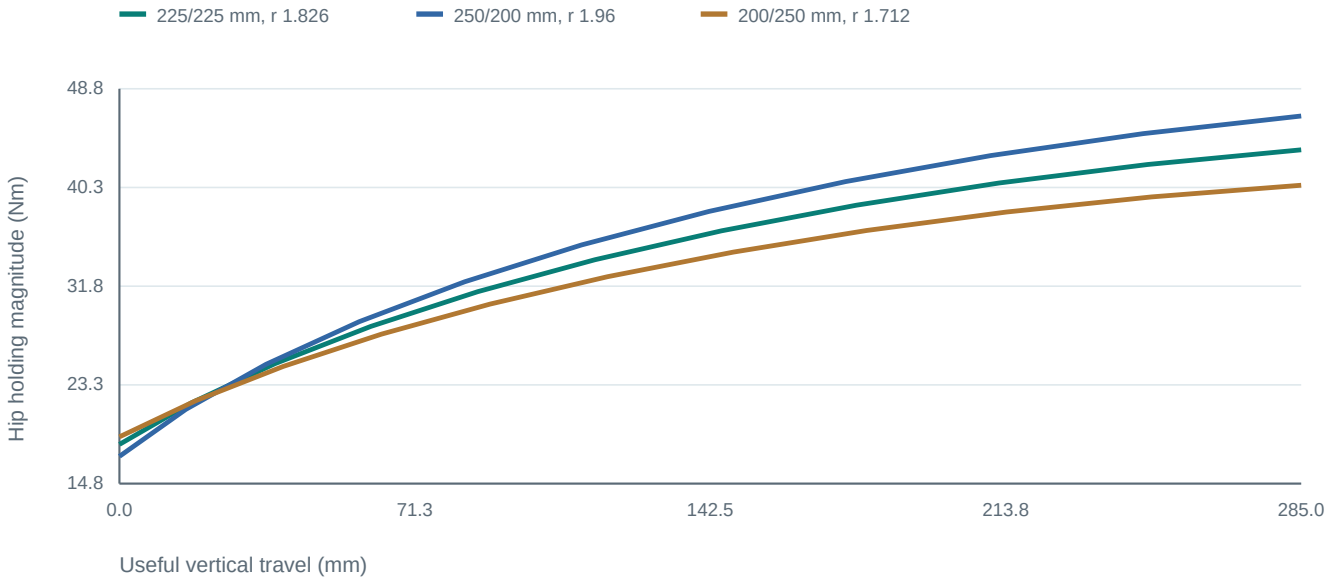
Reverse lengths can lower hip RMS

Matched 450 mm total link length, exactly +50 mm sideways movement and 325/285 mm height.

Lower link=1.25 x upper is the requested reverse orientation. At the same total size and starting angle, it gives lower hip-equivalent RMS after retuning the guide to the same lateral movement. It is not a universal motor benefit: equivalent relative-knee torque gives a different ordering.

Upper/lower mm	Guide ratio	Hip RMS Nm	Knee RMS Nm	Hip peak Nm
225/225	1.825724	35.433	19.408	43.584
250/200	1.959861	37.254	19.008	46.496
200/250	1.711531	33.508	19.578	40.535

Individual direct-hip holding torque over the useful stroke



Reverse 200/250 gives **5.43% less hip RMS** than equal 225/225 and 10.06% less than upper-long 250/200. Both endpoints are solved for the same 325 mm vertical task; the normal end angles differ (165.613 /164.802 /167.577 degrees). Guide ratio changes are part of this design comparison.

All use alpha=phase=0, start100 degrees, 20+20 mm assumed reserves and 110.3625 N vertical per leg (45 kg shared over four corners). Moving links are massless; no spring, inertia or loss is included. RMS here is uniform useful height, equivalent to constant vertical speed without dwell. These are not motor ratings.

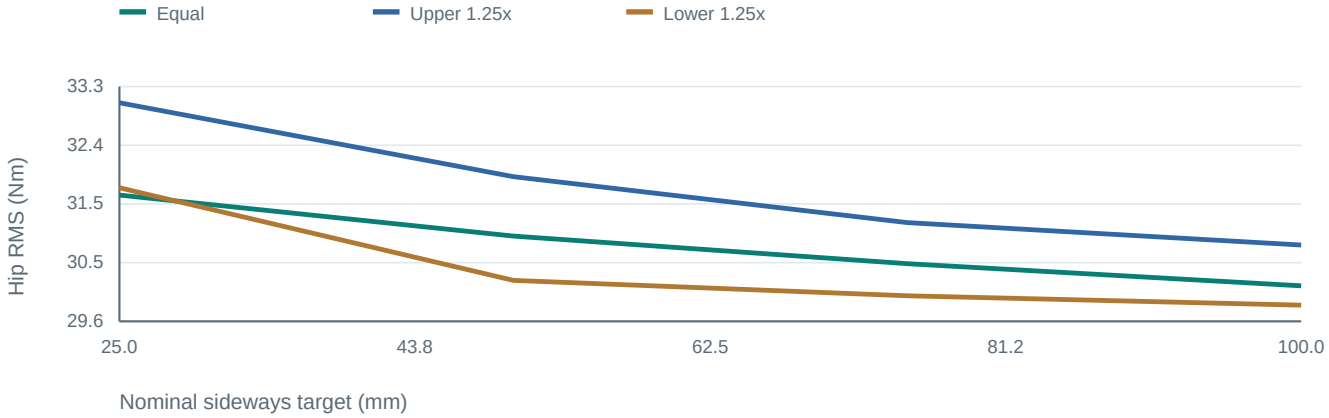
02 / RMS COMPARISON

Compare compact designs fairly

Bounded RMS grid leaders for approximately 50 mm sideways movement; equal links remain eligible.

Angle cap deg	Family	Upper/lower mm	Guide	Hip RMS Nm
165	Equal	230.0/230.0	1.82	35.906
165	Upper 1.25x	250.0/200.0	1.96	37.256
165	Lower 1.25x	211.1/263.9	1.71	34.822
170	Equal	207.5/207.5	1.81	33.262
170	Upper 1.25x	225.0/180.0	1.94	34.434
170	Lower 1.25x	191.1/238.9	1.71	32.389
175	Equal	190.0/190.0	1.79	30.952
175	Upper 1.25x	202.8/162.2	1.93	31.878
175	Lower 1.25x	175.6/219.4	1.70	30.262

175-degree cap: best retained grid hip RMS in each lateral band



At 175 degrees the lower-long 175.6/219.4 mm, r1.70 gives 30.262 Nm, versus 30.952 Nm for equal190/190,r1.79: about2.23% lower. However, its largest link is15.5% longer and combined length3.95% larger. These leaders occupy different +/-2.5 mm lateral bands (48.06 versus52.36 mm), so use the exact50 mm fixed-size comparison on page1 to isolate a common lateral task.

The reverse family has the lowest retained hip RMS in the50 mm band at each cap. It does not win every case: equal links slightly beat reverse at25 mm and cap175. Results depend on path, guide, size limits and operating range; grid leaders are not global continuous optima.

03 / RMS COMPARISON

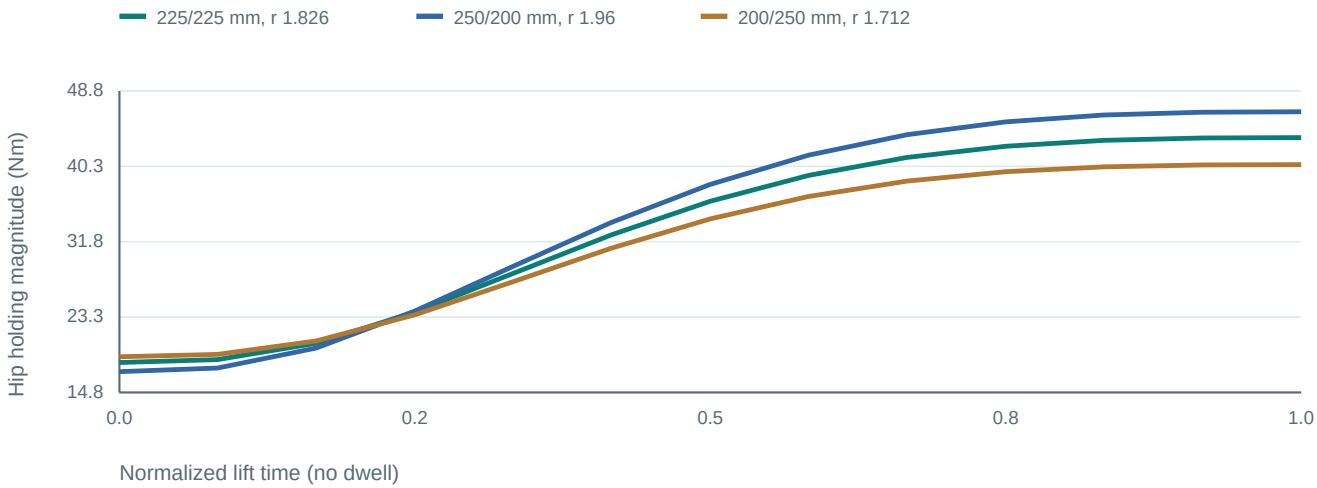
RMS depends on the actuator and duty

Static load screening must distinguish hip, relative knee and the motor shaft.

At guide2:1, fixed S=L2+L3 gives Jz=S cos(q-pi), independent of length difference. **Equal, upper-long and lower-long links therefore have identical vertical-load RMS, peak, ROM and height history.** Their sideways movement differs.

Upper/lower mm, r2	Hip RMS Nm	Knee RMS Nm	Signed lateral mm
225/225	37.743	18.872	0.000
250/200	37.743	18.872	39.563
200/250	37.743	18.872	-39.563

Same smooth quintic height motion: static hip holding torque only



For page1 cases, static RMS over the common smooth quintic lift is 34.734 /36.497 /32.856 Nm (equal /upper-long /lower-long); the reverse advantage persists for this assumed height history. No acceleration torque is included. Uniform height RMS is not the same weighting as uniform joint angle or smooth time.

Coordinate mapping: tau_knee=-tau_hip/r; knee speed=-r times hip speed. For page1, knee RMS is19.408 /19.008 /19.578 Nm. Reverse is about0.88% higher than equal in that coordinate. A separate motor gearbox changes torque and speed again. Compare the actual actuator location, gearing, inertia and duty before choosing.

Ideal useful lift work remains31.453 J. RMS helps estimate copper-loss heating only for a declared motor/gear/duty; it does not establish energy efficiency. Spring compensation, moving masses, side loads and dwell can change all rankings.

04 / RMS COMPARISON

Useful reverse examples to examine

Both keep lower/upper=1.25, alpha=phase=0 and the existing mathematical height inverse.

Upper/lower mm	Guide	Sideways mm	Hip RMS Nm	End angle deg
180/225	1.70000	49.10	30.835	173.402
200/250	1.71429	49.20	33.550	167.519

180/225 mm with17:10 is a rounded compact reverse example:49.10 mm sideways travel,30.835 Nm hip RMS and36.814 Nm peak. It slightly improves on the earlier equal190/190,r1.8 example (31.078 Nm RMS), but uses a longer individual link and25 mm more combined length. 200/250 with12:7 is a larger rational-guide example, not a lower-torque upgrade to the compact case.

Physical inversion remains unresolved. At the reverse fold, C lies beyond A by L3-L2 while the lower link passes through A in side projection. A45-50 mm offset does not clear a200 mm wheel or prove the belt/chassis layout. The screen only assumes rb60 mm and10 mm pitch-circle surplus; outer hardware, axial stack, tire, fork, chassis and structural dimensions are unknown.

Search:582,471 combinations; S350-600 by5 mm, L2/L3=1/1.25/0.8, guide1.2-2.6 by0.01, starts100-140 by5 degrees, separate caps165/170/175, 325 mm height,20+20 reserves, lateral25/50/75/100 +/-2.5 mm, monotone normal sideways motion and full100-260 height inverse with Jz>=5 mm/rad. 36 selected grid leaders are checked continuously for extrema after sampled screening. Only hip RMS is optimized; knee and quintic values are evaluations, not separate optima.

Validation: 52 analytical cases; 7,702 independent originalV2/Andy comparisons pass. Height RMS error 4.54e-12 Nm; quintic RMS error 9.89e-12 Nm. Independent adaptive quadrature and doubled time resolution also pass. Three actual positive-mass Pymunk full100-260 degree sweeps pass at2/1/0.5/0.25 ms and converge; finest path errors0.00755/0.00847/0.00736 mm. Zero gravity/forces/collision shapes: geometry only. No loaded Pymunk, spring, impact, contact, clearance, hardware or thermal validation is claimed.

Keep all earlier reports, model decisions and the dual-rope IMPOSSIBLE/rejected status. Refit springs and actual motor dynamics only after selecting geometry, physical packaging and duty. Source c92880d; isolated workspace; canonical physics NOT APPLICABLE. Reproduction commands and evidence are in the adjacent design record.